

Predicting Injury Severity of Work Zone Crashes Along Florida Freeways

Transportation Research Record
2026, Vol. 2680(1) 550–567
© The Author(s) 2025
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/03611981251357938
journals.sagepub.com/home/trr



Hellen Shita¹ , HM Nayem¹ , and Priyanka Alluri¹ 

Abstract

With increasing demand for capacity improvement, future highway construction, and the need for newer infrastructure, the United States is likely to experience more, longer duration, and longer stretches of work zones along its roadways. Work zones are often characterized by lower speeds, changes in traffic patterns, and sometimes narrower lanes, leading to safety compromises of workers, motorists, and other road users. As higher speed limits usually characterize freeways, the significant speed reductions along work zones create substantial shifts for drivers. The abrupt changes in speed and traffic patterns influence the occurrence of crashes as well as injury severity. This study used 2018 to 2022 Florida work zone data in conjunction with freeway crashes to determine the factors influencing the injury severity of crashes occurring along its freeway work zones. Six models, including the K-nearest neighbor, support vector machine, random forest, extreme gradient boosting, multinomial logistic regression, and ordinal logistic regression, were compared for prediction performance using various metrics. The random forest model was observed to be superior in its classification capacity and therefore used for the data analysis. Variable importance and partial dependency plots were used to interpret the model and understand the influence of roadway, environmental, temporal, and human factors associated with work zone crash injury severity. The findings from this study will be helpful to traffic engineers and other responsible agencies in freeway work zone planning to reduce crash injuries and improve roadway safety.

Keywords

operations, work zone, safety, crash analysis, crash data, crash severity

According to the *Manual on Uniform Traffic Control Devices*, work zones are areas along roadways where construction, maintenance, or utility activities take place (1). Work zone areas are usually marked by signs, closed lanes, channelizing devices, barriers, pavement markings, and work vehicles (2, 3). Work zones are characterized by four major component areas: the warning area, the transition area (drivers channelized to a new path), the activity area (actual construction area), and the termination area (used to return drivers to their original path) (4). Work zones play a crucial role in improving existing roadways through new construction and maintenance activities, including repainting, resurfacing, and pothole repairs. These improvements lead to enhanced safety and mobility as well-maintained roads reduce crash risks, improve traffic flow, and extend the lifespan of transportation infrastructure, supporting safe and more reliable travel experiences for all.

Even though work zones are inevitable, they pose both safety and mobility challenges for motorists and workers. The presence of equipment, work vehicles, signs, and channelizing devices can affect driver behavior. Previous studies have established a correlation between variations in lane width and increased driver stress levels (5). This complex navigation along work zones, accompanied by changes in lane widths, usually results in confusion and frustration (6) requiring heightened caution, making navigation through work zones more complex. In other words, the driving complexity along work zones often leads to unsafe driving behaviors

¹Department of Civil and Environmental Engineering, Florida International University, Miami, FL

Corresponding Author:
Hellen Shita, hshit001@fiu.edu

(6) including distraction, speeding, sudden lane changes, and abrupt speed reduction, which compromise the safety of both drivers and workers. According to the Fatality Analysis Reporting System, distraction contributed to 11.7% of fatal work zone crashes, whereas it led to 7.8% of fatal non-work-zone crashes between 2020 and 2022 (7). Sudden lane changes contribute to the occurrence of sideswipe crashes whereas abrupt speed reductions tend to result in rear-end crashes along work zones. The temporary traffic control management design, workmanship, and quality and frequency of inspections also contribute to the occurrence of collisions and delays along work zones. Previous studies have also identified that work zones result in about 24% of nonrecurring delays and 10% of total delays (8). These delays lead to traffic congestion, which not only affects motorists, but also slows down the region's economy (9).

The National Work Zone Safety Information Clearinghouse reports about 108,000 work zone crashes occurring annually in the United States, ranging from 94,000 to 123,000 (10). Among those, about 24,000 crashes result in injuries whereas about 52,000 result in property damage only. These crashes include, on average, more than 700 fatal crashes and 800 fatalities every year along work zones (7). These statistics highlight the potential safety risks associated with the presence of work zones. Despite the efforts by various agencies to reduce crashes and injury severity, work zone crashes still occur. The occurrence of crashes may be the result of the difficulty in simultaneously ensuring traffic flow, road safety, and roadwork progress. This difficulty is further exacerbated in high-speed facilities, such as freeways (11). Work zones on freeways present unique and heightened safety challenges because of higher speed limits, increased traffic volumes (12), and increased braking distances (13). Owing to higher speeds on freeways, any disruption is more hazardous for motorists and road workers compared with low-speed facilities. Although the number of work zones is likely to increase, it is essential to understand how safety in these critical areas can be improved, by understanding the factors affecting the injury severity and developing targeted countermeasures.

Literature Review

Ensuring road user safety, worker safety, and roadwork progress simultaneously may be challenging, especially on high-speed facilities (11). Several studies have confirmed the considerable effect of work zones on safety. In Italy, La Torre et al. found that fatal and injury crashes increased by about 33% and property damage only (PDO) crashes increased by 66% along work zones (14). In Florida, Xie et al. identified that work zone crashes

are prone to severe injuries compared with other crashes (15). These statistics raise the need for more research on work zone crashes and emphasize the need to understand the specific factors attributing to these incidents.

Multiple studies have investigated the factors attributing to injury severity among work zone crashes. In New Zealand, Sze and Song identified that crashes involving motorcyclists, pedestrians, or workers along work zones were associated with severe injuries compared with those that did not (16). A possible explanation for the trend was the lack of physical separation between vehicles and vulnerable road users (16).

Zhang and Hassan identified that various factors influence the severity of work zone crash injuries differently during daytime and nighttime conditions (17). Using data from work zones in Egypt, they identified that younger drivers (<35 years), older drivers (>50 years), and male drivers influence injury severity during nighttime compared with middle-aged drivers and female drivers. The authors attributed their findings to inexperienced driving among young drivers and the longer reaction times among older drivers. The increased severity of the problem with male drivers was attributed to male drivers' higher tendency to take risks, such as speeding and driving aggressively.

Ghasemzadeh and Ahmed identified that weather and lighting conditions are among the most critical factors determining the severity of work zone crashes (18). Their study indicated that drivers involved in work zone crashes at dusk and dawn were highly likely to be involved in severe crashes. Using Minnesota work zone crash data, Osman et al. identified that partial control of access, rural roadways, evening and weekend times, and curved segments were associated with increased injury severity (19). Several other studies have identified factors attributing to different injury severities among work zone crashes, including those related to roadway, environmental, human, vehicle, and temporal factors (20–23). However, research on freeway work zone safety remains limited, despite the role of freeways in traffic mobility, connectivity, economic development, and recreational travel (24).

To evaluate the safety impact of work zones on freeways, this study integrates historical work zone data and crash data to identify the factors affecting injury severity in freeway work zone crashes. Crashes were matched with historical freeway work zone data from the Florida Department of Transportation (FDOT) work zone archive, using date and milepost of occurrence for the 5-year analysis period, from 2018 to 2022. This approach enhances the accuracy and relevance of the analysis and has the potential to offer valuable insights on Florida's work zone safety performance. Findings from this study could help transportation agencies better understand the

Table 1. Distribution of Work Zone Segments by Roadway and Work Zone Characteristics

Characteristic	2018		2019		2020		2021		2022		Avg. no. of segments	Avg. ft/day
	No.	ft/day	No.	ft/day	No.	ft/day	No.	ft/day	No.	ft/day		
Work description												
Construction	7	82.2	32	94.3	45	51.2	44	67.5	55	39.8	37	67.0
Maintenance	2	5.19	6	79.2	10	23.3	17	18.70	20	53.8	11	36.1
Resurfacing	10	134.2	30	132.1	42	103.3	47	88.7	64	78.9	39	107.5
Traffic operations	3	6.9	6	64.5	10	34.4	18	29.9	17	27.7	11	32.8
Speed limit (mph)												
25–45	2	4.9	5	4.3	4	2.3	4	7.6	9	8.9	5	5.7
50–55	1	2.2	2	14.4	5	48.9	7	47.0	12	23.9	5	27.3
60–65	3	48.1	25	48.0	44	31.2	46	29.5	43	29.7	32	37.3
70	16	127.5	42	149.3	54	108.2	69	97.3	92	79.5	55	112.4
Functional class												
Expressway rural	4	164.6	24	253.5	9	114.5	13	67.7	16	83.5	13	136.8
Interstate rural	5	54.9	59	105.6	24	103.5	20	121.8	31	94.7	28	96.1
Expressway urban	6	122.4	86	91.0	37	51.6	44	51.5	50	42.1	45	71.7
Interstate urban	7	62.2	91	65.5	37	54.4	49	58.3	59	44.0	49	56.9

Note: Avg. = average.

safety status of freeway work zones and help implement targeted countermeasures.

Data Collection and Description

This section explains the data used, the data collection process, and the descriptive statistics of the data used for analysis. The work zone data were obtained from FDOT with 1,688 observations, including work zone descriptions, beginning and end dates, and beginning and end mileposts. Using Florida's Roadway Characteristics Inventory, freeways were identified as those with functional interstate and expressway classifications in rural and urban settings. Spatial join analysis in ArcGIS was used to identify work zones along the freeways. A total of 261 work zone segments were identified on the freeways. Table 1 shows the distribution of work zone exposure by roadway and work zone characteristics, showing the number of segments and the feet per day (ft/day). The ft/day values were calculated by dividing the total number of miles by the total number of days of the corresponding characteristic and multiplying by 5,280 to convert miles into feet for easy reference. The roadway characteristics include the type of work zone activity, functional classification with the nature of the area, and the aggregate speed limits along the freeway work zones.

The statistics indicated that resurfacing activities accounted for the highest number of work zone mile-days, averaging 107.5 ft/day. Resurfacing activities were significantly more prevalent than construction activities, which averaged 67 ft/day. With regard to speed limits, a greater number of work zone activities occurred on roadways with a 70-mph limit, although posted speed limits

often varied depending on the specific work zone conditions. Although some zones stipulated reduced speed limits for safety, others maintained the existing limits. In addition, rural expressways recorded the highest average work zone mileage (136.8 ft/day), followed by rural interstates (96.1 ft/day). Overall, rural areas had more identified work zone segments compared with urban areas.

Work zone crash data were obtained from the Signal Four Analytics database. The database includes various crash information, including injury severity, and environmental, human, roadway, temporal, and vehicle characteristics. Injury severity is identified using the KABCO scale (K = fatal injury, A = incapacitating injury, B = nonincapacitating injury, C = possible injury, O = no injury/property damage only).

According to the Signal Four Analytics database, a total of 14,947 work zone crashes occurred along Florida freeways between 2018 and 2022. However, using the temporal correlation of the work zone crashes with the work zone data from the FDOT work zone archive, only 1,258 crashes were mapped with respective work zones, based on their spatial location and date of occurrence. Only crashes identified within the work zone begin and end mileposts and within the work zone beginning and end dates were included in the analysis (i.e., 1,258 crashes). This approach was used to identify the influence of the type of work zone activity (construction, maintenance, traffic operations) on injury severity, a feature that was only available on the work zone archive. The approach would also ensure that all crashes included in the analysis actually occurred along work zones.

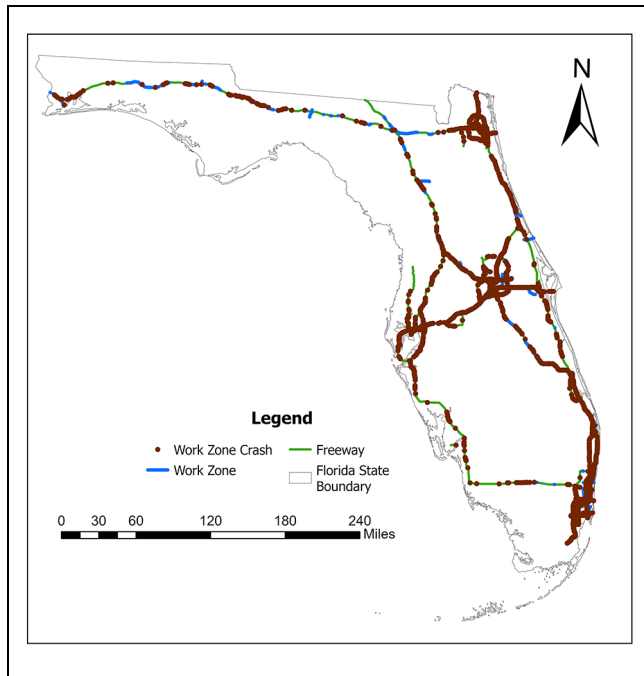


Figure 1. Florida map showing freeway work zones and crashes (2018 to 2022).

However, this process revealed a significant discrepancy in the number of identified work zone crashes. This inconsistency is likely to be a result of the limited availability of comprehensive work zone data. Although the Signal Four Analytics crash database reported a larger number of work zone crashes, the work zone archive lacked sufficient data to fully account for all the

crashes recorded in the Signal Four Analytics database. This limitation would pose a challenge for crash frequency analysis, which requires complete data coverage to accurately analyze the total number of crashes. However, for crash injury severity analysis, full data inclusion is not required as long as the missing data are randomly distributed and not biased. Therefore, the absence of the missing data did not introduce bias nor compromise the validity of the injury severity findings. Figure 1 presents a map showing the distribution of work zones and crashes along Florida freeways between 2018 and 2022.

Figure 2 shows the normalized freeway work zone crash distribution by year and injury severity. The normalization was done per mile-day for each particular year. For instance, if a work zone spans 3 years, its total number of days is distributed accordingly for each year. The miles were consistent across all years, provided the work zone was assumed to be active between the begin and end date. Figure 2 shows the maximum value observed in 2018, followed by 2021. The lowest crash frequency was observed in 2019.

Descriptive Statistics

This section summarizes the descriptive statistics of the crash data used in this analysis. The data were grouped into temporal, roadway, and environmental factors, driver characteristics and vehicle factors. The data were classified into three injury classes, with fatal and incapacitating injuries combined (K + A), nonincapacitating and possible injury combined (B + C), and no injury

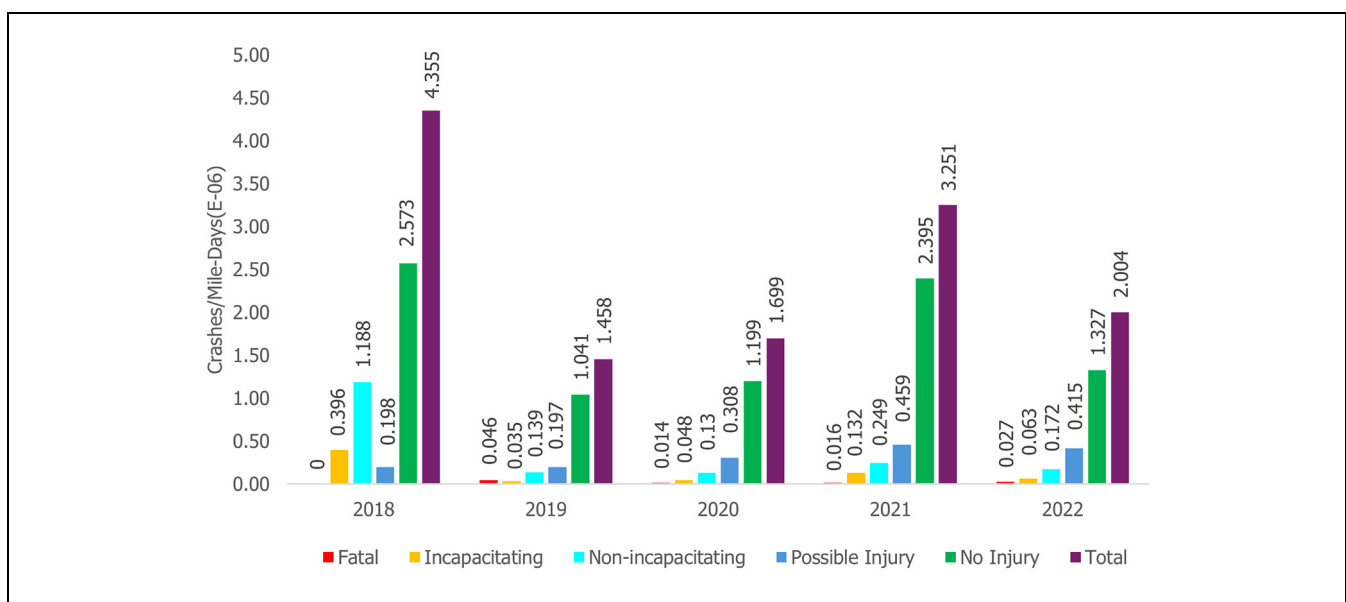


Figure 2. Freeway work zone crashes distribution by year and crash severity in Florida.

(PDO). Table 2 shows the descriptive statistics of the data.

The statistics showed that most crashes occurred in the evening off-peak hours ($n = 286$) followed by day off-peak hours ($n = 277$). However, the largest proportion of K + A crashes were observed during late-night hours (7.8%) followed by morning peak hours (6.3%). Similarly, a higher proportion of K + A crashes was observed for crashes that occurred on weekends (5.7%) compared with 4.2% that occurred on weekdays. More work zone crashes occurred in urban areas and along interstates compared with rural areas and on expressways. Freeway crashes occurring in construction work zones were predominant. However, a larger percentage (6.3%) of those that occurred in resurfacing activities resulted in fatalities or serious injuries. In addition, more crashes were observed to occur along the activity area than in other areas in a work zone. More than 14% of work zone crashes along the termination area in freeway work zones resulted in fatal and serious injuries. More than half of the crashes occurred while workers were absent. However, as expected, those that occurred in the presence of workers were observed to be more severe.

Methodology

Variable Correlation

This section discusses the steps and algorithms performed to identify the factors influencing crash injury severity in freeway work zones in Florida. A correlation matrix was created using Pearson correlation coefficients to identify the variables that were highly correlated to each other. A threshold value 0.5 was used to identify highly correlated variables (25–27). Figure 3 shows the correlation matrix as obtained from the “cor” function in the R programming language. Aggressive driving was found to be correlated with speeding, with a coefficient of 0.93, and was therefore removed from the analysis. The speeding variable was retained in the model for further analysis.

Boruta Algorithm

For practical reasons, selecting a small and minimal feature set is always desirable (28). The Boruta algorithm in R, suitable for any regression or classification problem (29), was utilized for feature selection, as it identifies relevant features with respect to the dependent or target variable (crash injury severity in this case). The algorithm has been commonly used for feature selection in several fields, including traffic safety, medicine, and social economics (30, 31). Feature selection allows for reduced model computation time, improved accuracy, and network overhead reduction (29). The Boruta algorithm results illustrated in Figure 4 confirmed 15 strong

impact variables (in green), 3 tentative (in yellow), and 5 rejected variables (in red). The blue-colored variables indicated the minimum, mean, and maximum values of the shadow features, which served as a benchmark for evaluating the relevance of actual variables. Note that only the confirmed (green) and tentative (yellow) variables were used in the analysis. The rejected variables (red) were omitted from further analysis, that is, interchange location, vehicle age, roadway alignment, driver's gender, and nature of the area where the crash occurred (i.e., rural or urban).

Classification

To ensure the best-performing model was used, six models were run and their performance metrics compared. The six models included K-nearest neighbor (K-NN), support vector machine (SVM), random forest (RF), extreme gradient boosting (XGBoost), multinomial logistic regression (MLR), and ordinal logistic regression (OLR). These models were selected for their proven ability to classify categorical crash data effectively and their frequent use in previous safety studies. Selecting the most suitable model is essential to achieve optimal model fit and produce reliable, accurate results. The following section explains the algorithms and assumptions of machine learning and statistical models used for classification.

K-Nearest Neighbor. The K-NN method is a supervised machine learning strategy commonly utilized in traffic and transportation (32). This technique classifies observations based on proximity to other observations using the nearest neighbor decision rule. The majority of the closest observations determine the class of the observation. Two crucial decisions in the K-NN method are the choice of “k” and the distance function, represented by Equation 1.

$$D_e = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (1)$$

The optimal “k” value is usually found through experimentation to maximize forecast accuracy. The K-NN distance function is based on the Euclidean distance between two points in two dimensions (33).

Random Forest. RF is a robust machine learning technique that addresses overfitting by creating multiple decision trees using different predictor variables (34). RF is a widely utilized ensemble learning method in transportation safety research (35, 36). RF constructs an ensemble of decision trees by altering the training set using the bootstrap sampling method. In this study, Equation 2 represents the input for RF, whereas the output pertains to the injury severity of an individual sample (37).

Table 2. Distribution of Work Zone Crashes by Injury Severity

Variable	Category	K + A		B + C		PDO		Total
		n	%	n	%	n	%	
Temporal characteristics	Time of crash							
	Morning peak (6:00–9:59 a.m.)	16	6.30	57	22.50	180	71.10	253
	Day off-peak (10:00 a.m.–2:59 p.m.)	12	4.30	65	23.50	200	72.20	277
	Evening off-peak (7:00–11:59 p.m.)	13	4.50	89	31.10	184	64.30	286
	Evening peak (3:00–6:59 p.m.)	3	1.10	62	22.50	210	76.40	275
Day of the week	Late nighttime (12:00–5:59 a.m.)	13	7.80	48	28.70	106	63.50	167
	Weekday	43	4.20	246	24.30	725	71.50	1,014
	Weekend	14	5.70	75	30.70	155	63.50	244
Roadway characteristics	Location							
	Rural	26	5.00	126	24.20	369	70.80	521
	Urban	31	4.20	195	26.50	511	69.30	737
	Expressway	16	4.50	75	21.30	261	74.10	352
Interchange related	Interstate	41	4.50	246	27.20	619	68.30	906
	No	51	4.30	300	25.60	822	70.10	1,173
	Yes	6	7.10	21	24.70	58	68.20	85
Work zone description	Construction other than mill or resurface	21	3.60	144	24.70	418	71.70	583
	Maintenance	5	3.20	44	28.60	105	68.20	154
	Resurfacing	24	6.30	91	23.90	265	69.70	380
	Traffic operations	7	5.00	42	29.80	92	65.20	141
	Activity area	38	4.20	234	25.70	637	70.10	909
	Advance warning area	8	4.80	38	22.80	121	72.50	167
	Termination area	3	14.30	5	23.80	13	61.90	21
Workers present	Transition area	8	5.00	44	27.30	109	67.70	161
	No	28	3.70	190	24.90	545	71.40	763
Lane departure	Yes	29	5.90	131	26.50	335	67.70	495
	No	14	2.40	149	25.30	425	72.30	588
Horizontal alignment	Yes	43	6.40	172	25.70	455	67.90	670
	Straight	52	4.60	296	26.10	787	69.30	1,135
	Curve	5	4.10	25	20.30	93	75.60	123
Vertical alignment	Level	50	4.30	289	25.00	816	70.70	1,155
	Grade	7	6.80	32	31.10	64	62.10	103
Type of impact	Angle	7	11.90	22	37.30	30	50.80	59
	Fixed object	27	6.00	94	20.80	331	73.20	452
	Rear end	17	3.30	161	31.70	330	65.00	508
Lighting conditions	Sideswipe	6	2.50	44	18.40	189	79.10	239
	Dark: lighted	14	4.70	86	28.80	199	66.60	299
	Dark: not lighted	13	7.70	44	26.20	111	66.10	168
	Daylight	28	3.90	171	23.60	525	72.50	724
Weather condition	Dusk/dawn	2	3.00	20	29.90	45	67.20	67
	Clear	41	4.70	229	26.50	594	68.80	864
	Cloudy	12	6.40	41	21.90	134	71.70	187
	Severe weather	4	1.90	51	24.60	152	73.40	207

(continued)

Table 2. (continued)

Variable	Category	K + A		B + C		PDO		Total
		n	%	n	%	n	%	
Number of vehicles involved	1 vehicle	23	6.10	78	20.80	274	73.10	375
	2 vehicles	25	3.40	181	24.30	538	72.30	744
	≥ 3 vehicles	9	6.50	62	44.60	68	48.90	139
Driver characteristics	Female	14	3.60	99	25.40	277	71.00	390
	Male	43	5.00	222	25.60	603	69.50	868
	≤ 21 years	6	3.40	44	24.70	128	71.90	178
Driver's age	22–34 years	21	4.50	137	29.30	310	66.20	468
	35–64 years	29	5.50	123	23.10	380	71.40	532
	≥ 65 years	1	1.30	17	21.30	62	77.50	80
Aggressive	No	55	4.60	306	25.60	834	69.80	1,195
	Yes	2	3.20	15	23.80	46	73.00	63
DUI	No	48	3.90	310	25.30	868	70.80	1,226
	Yes	9	28.10	11	34.40	12	37.50	32
Distraction	No	43	3.90	265	24.20	786	71.80	1,094
	Yes	14	8.50	56	34.10	94	57.30	164
Speeding	No	57	4.70	307	25.50	839	69.70	1,203
	Yes		0.00	14	25.50	41	74.50	55
Speed limit	25–45 mph	2	6.90	9	31.00	18	62.10	29
	50–55 mph	11	5.40	59	29.10	133	65.50	203
	60–65 mph	23	3.40	163	24.10	490	72.50	676
	70 mph	21	6.00	90	25.70	239	68.30	350
Vehicle characteristics	Motorcycle	4	22.20	8	44.40	6	33.30	18
	Passenger car	45	4.20	282	26.00	757	69.80	1,084
	Heavy vehicle	8	5.10	31	19.90	117	75.00	156
Vehicle age	≤ 10 years	36	4.20	211	24.70	607	71.10	854
	> 10 years	21	5.20	110	27.20	273	67.60	404

Note: K + A = fatal and incapacitating injuries combined; B + C = nonincapacitating and possible injury combined; PDO = no injury (property damage only); DUI = driving under the influence.

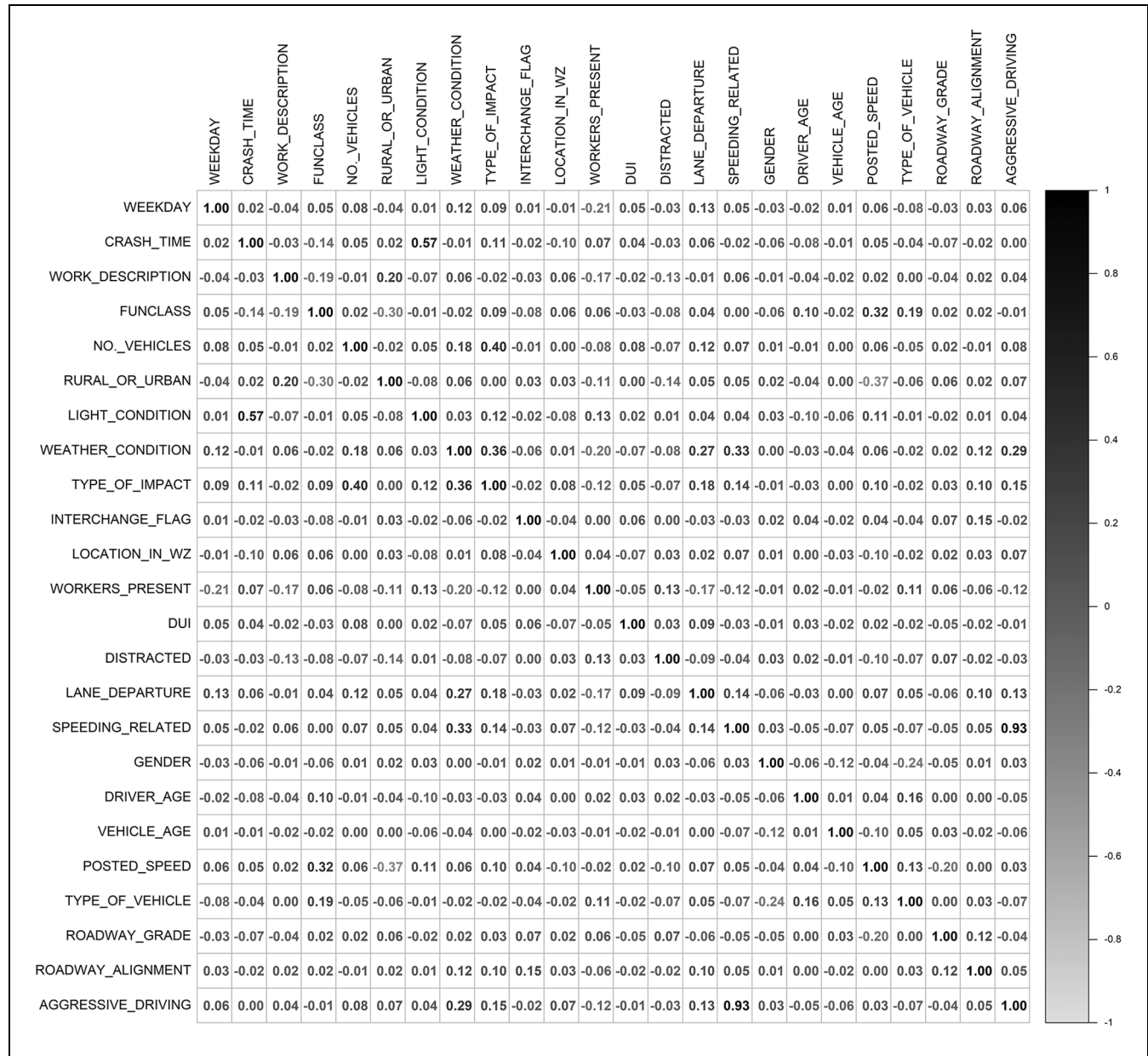


Figure 3. Pearson correlation matrix for independent variables.

Note: DUI = driving under the influence.

$$\{X = [x_{i1}, x_{i2}, \dots, x_{im}], y_i\} i = 1, 2, \dots, n; y_i \in \{1, 2, 3\} \quad (2)$$

where

X denotes data samples,

n represents number of data samples,

m represents number of variables, and

y represents injury severity.

The RF procedure involves selecting a sample database, generating decision trees, and merging decision trees. Each tree in the ensemble contributes a vote toward the final classification outcome, with the most common class

label among all trees being chosen as the final prediction (38).

Support Vector Machine. SVM's superior and comparable predictive performance has recently been documented in several crash injury severity studies (39, 40). The SVM draws an optimal separating hyperplane in this higher-dimensional space to separate the points into groups while maximizing the margin between the linear decision boundaries. In the context of SVM, the input is represented as vectors $x_i \in R^n$ for $i = 1, 2, \dots, N$, which represents the

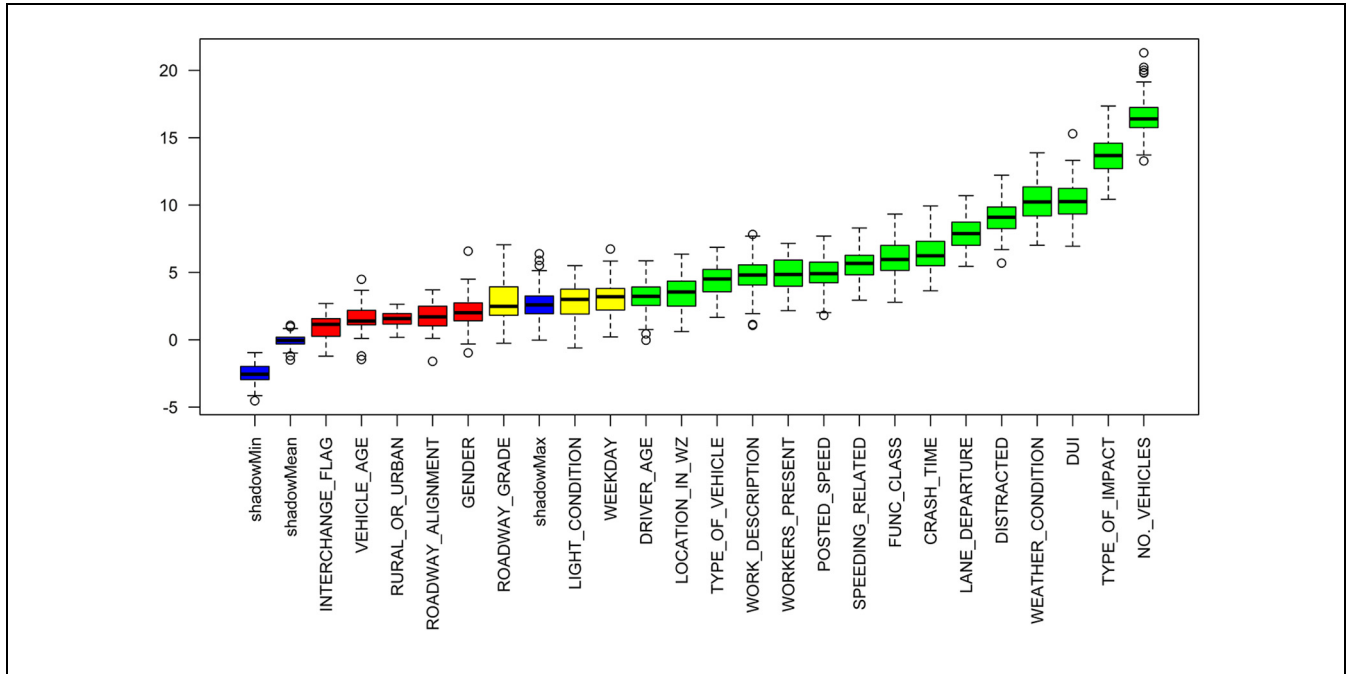


Figure 4. Boruta variable importance.

Note: DUI = driving under the influence.

full set of explanatory variables, and the training output is defined as $y_i \in R^n$, which represents the injury severity of crashes (33). The hyperplane for separating outcomes can be written as a set of points X satisfying

$$T \cdot X - b = 0 \quad (3)$$

where “.” denotes dot product, and vector T is a normal vector perpendicular to the hyperplane.

Extreme Gradient Boosting. Chen and Guestrin improved the original gradient boosting decision tree algorithm in 2016 (41). The decision rules of XGBoost are very similar to those of a decision tree. It can be used for regression as well as classification. Although the gradient boosting machine has recently been used in traffic and transportation, XGBoost is its more efficient and scalable version (42). The overall formulation of the model is provided in the following (43):

$$\hat{y}_i(x) = \sum_{T=1}^T f_T(x_i), (f_T \in F) \quad (4)$$

where $y_i(x)$ denotes the predicted value, T is the total number of trees, $f_T \in F$ represents the T -th regression tree selected from the functional space of all possible trees F . This equation is further optimized for the objective function,

$$\text{Obj}(\theta) = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{T=1}^T \Omega(f_T) \quad (5)$$

XGB introduces an incremental regularization term as follows:

$$\Omega(f_T) = \gamma t + \frac{1}{2} \lambda \sum_{j=1}^t C_{q(x)^j} \quad (6)$$

where

t is total number of leaves,

$C_{q(x)^j}$ is score for j^{th} leaf,

λ and γ are regularization parameters.

Multinomial Logistic Regression. If the dependent variable is Y , which has i th degrees, and independent variables are considered as $X_{i1}, X_{i2}, \dots, X_{ij}$ and j is the number of predictors, thus, the MLR model for the crash, k , having severity level, i , can be expressed as follows (44):

$$\begin{aligned} \text{Logit}(P_k(Y_{ij} \leq 1 | X_j)) &= \text{Ln} \left(\frac{P_k(Y_{ij} \leq 1 | X_j)}{1 - P_k(Y_{ij} \leq i | X_j)} \right) \\ &= \alpha_i + \beta_{i1}X_{i1} + \beta_{i2}X_{i2} + \dots \\ &\quad + \beta_{ij}X_{ij} \end{aligned} \quad (7)$$

where α_i , and β_{ij} , represent the constant for the crash severity level, i , and the regression coefficient, respectively. $P_k(Y_{ij} \leq i | X_j)$ is the cumulative probability, Y_{ij} , under the conditional form of $i | X_j$ and

Table 3. Summary of Model Prediction Performance Metrics

Model	AUROC			Overall AUROC	Overall prediction accuracy
	K + A	B + C	PDO		
K-NN	0.6619	0.5662	0.5949	0.5990	0.6896
SVM	0.6461	0.6460	0.6995	0.6834	0.7082
RF	0.6516	0.6726	0.7128	0.6998	0.7108
XGBoost	0.6103	0.6267	0.6634	0.6517	0.6710
MLR	0.6064	0.6668	0.6821	0.6748	0.6923
OLR	0.6830	0.6562	0.6825	0.6758	0.6976

Note: AUROC = area under the receiver operating characteristic; K + A = fatal and incapacitating injuries combined; B + C = nonincapacitating and possible injury combined; PDO = no injury (property damage only); K-NN = K-nearest neighbor; SVM = support vector machine; RF = random forest; XGBoost = extreme gradient boosting; MLR = multinomial logistic regression; OLR = ordinal logistic regression.

$\sum_{i=1}^I P_k(Y_{1j} \leq i | X_1) = 1$. Thus, the multinomial logistic probability model can be expressed as follows (45):

$$P_k(Y_{ij} \leq 1 | X_j) = \frac{\exp(\alpha_i + \sum_{i=1}^I \beta_{ij} X_{ij})}{1 + \exp(\alpha_i + \sum_{i=1}^I \beta_{ij} X_{ij})} \quad (8)$$

$i = 1, 2, \dots, 5; j = 1, 2, \dots, J$

Ordinal Logistic Regression. OLR is an extension of the binary logistic regression model designed to handle multiple categories. It is well-suited for analyzing scenarios in which the injury level of the outcome variable is ordinal in nature (46). The logit form of the proportional odds model is expressed by Equation 9,

$$\begin{aligned} \text{logit}[\pi_j(x)] &= \ln\left(\frac{\pi_j(x)}{1 - \pi_j(x)}\right) \\ &= \alpha_j + (-\beta_1 X_1 - \beta_2 X_2 - \dots - \beta_p X_p) \end{aligned} \quad (9)$$

where $\pi_j(x) = P(Y \leq j | x_1, x_2, \dots, x_p)$, which gives the probability of being equal to or below the j level (level of injury severity in this context) for any given set of predictors, j is a discrete index $\{1, 2, \dots, j-1\}$, α_j is the cutoff point, and $\beta_1, \beta_2, \dots, \beta_p$ are the logit coefficients. To calculate the probability of being equal to or below the j^{th} level, Equation 9 can be rewritten as,

$$\begin{aligned} \text{logit}[P(Y \leq j | x_1, x_2, \dots, x_p)] &= \left(\frac{P(Y \leq \beta x_1, x_2, \dots, x_p)}{P(Y > x_1, x_2, \dots, x_p)} \right) \\ &= \alpha_j + (-\beta_1 X_1 - \beta_2 X_2 - \dots - \beta_p X_p) \end{aligned} \quad (10)$$

Receiver Operating Characteristic Curve

The ROC curve was used to assess the predictive abilities of the models. The ROC curve is a plot of sensitivity

versus specificity corresponding to different cutoff values of a predictor (47). The sensitivity of a predictor is the proportion of its true positives, whereas the specificity is the proportion of its false positives (47, 48). For a random guess, the ROC curve is the diagonal of the sensitivity versus specificity plot, equal to a value of 0.5. A perfect predictor would have an area under the curve (AUC) equal to 1.0, covering the whole plot from (0,0) to (1,1) (48). Table 3 summarizes the models' performance metrics, showing the AUC values of all the models with respect to the three classes in the target variable (K + A, B + C, PDO). In addition, the overall AUC values of the models and the overall prediction accuracy are also presented. Table 3 indicates that the RF model exhibited the highest AUC value in predicting the B + C and PDO classes whereas the OLR showed a higher AUC value in predicting the K + A class. However, the RF model had the highest overall AUC value, indicating higher prediction performance. Equation 11 shows the computation of the overall area under ROC curve.

$$\begin{aligned} \text{Overall AUROC} \\ &= \text{sum}(\text{auroc_values} \times \text{class_proportions}) \end{aligned} \quad (11)$$

In addition, the overall accuracy of the models was computed. The overall accuracy is the total number of correctly classified observations, calculated from the confusion matrix, and is defined as,

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (12)$$

where

TP is true positive,

TN is true negative,

FP is false positive, and

FN is false negative.

Accuracy provides the model's overall measure in correctly predicting the entire dataset. The RF model also

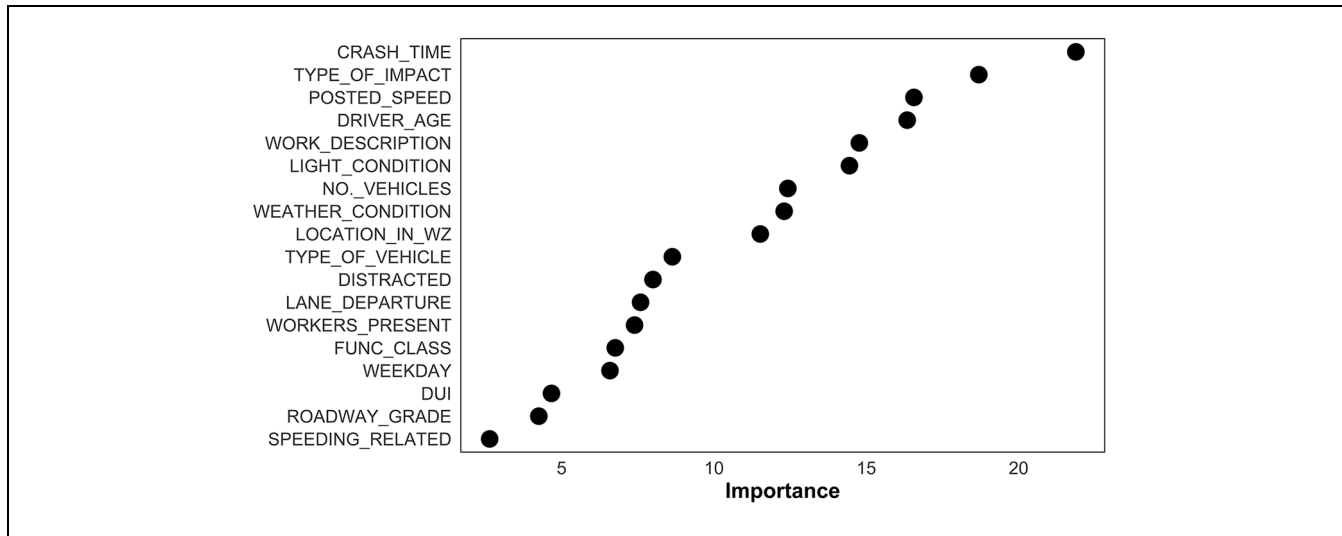


Figure 5. Variable importance for the random forest model (Gini index).

Note: WZ = work zone; DUI = driving under the influence.

exhibited the highest overall accuracy and was therefore utilized for further data analysis to identify crash severity influencing factors.

Results and Discussion

Variable Importance

The mean-importance values of each class were computed in *R* to obtain the overall importance value for each variable. The algorithm permutes the values of each variable and evaluates the resulting decrease in the scoring metric, Gini index for this case (49), using the RF model.

According to Figure 5, the time of crash occurrence had the highest contribution to the model's predictive performance. Type of impact, posted speed limit, and driver age ranked as the second, third, and fourth most important features influencing work zone crash severity, respectively. Figure 5 also indicates that the type of work zone activity (i.e., work description), lighting condition at the time of the crash, and the number of vehicles involved in a crash were very important features that contributed to the injury severity of freeway work zone crashes. However, the variable importance only exhibited variables and their contributing impact in the model's prediction. Note that the variable importance algorithm does not identify the effect of the variable categories in influencing the target variable.

As the results from a machine learning model are not directly interpretable, partial dependence plots (PDPs) were used to identify the influence of the predictor variables on injury severity. PDPs give a visual and graphical presentation of how independent variables affect the

crash injury severity (50). Other studies have utilized PDPs in interpreting machine learning models (51). The PDP presents the change in the average predicted value as the particular input(s) changes over their marginal distribution (52).

In other words, the partial dependence function estimates the average trend of a predictor variable against the probability of vehicle occupant injury occurrence for every value of the predictor variable after accounting for the average impacts of all other input variables in the model (53). The PDPs for categorical variables in this study were centered at a particular point of a variable to provide a simple comparison among the categories (53). The centering was obtained from the computation of the difference in the average prediction of each group from a base group, as shown in Equation 13,

$$\text{Marginal effect} = \left(\frac{P_{x_1} - P_{x_0}}{P_{x_0}} \right) \times 100 \% \quad (13)$$

where the marginal effect represents the impact of the predictor variable on the resulting injury severity, with x_0 as the base condition. For multiclass variables, the marginal effect was computed for each class/category, with one of the categories being the base category. Figure 6 presents the partial dependency plots for the K + A class since fatal and incapacitating injuries have more serious impacts.

Temporal Factors

The findings showed that crashes occurring during weekends had a higher probability of resulting in fatal and severe injuries compared with crashes occurring during

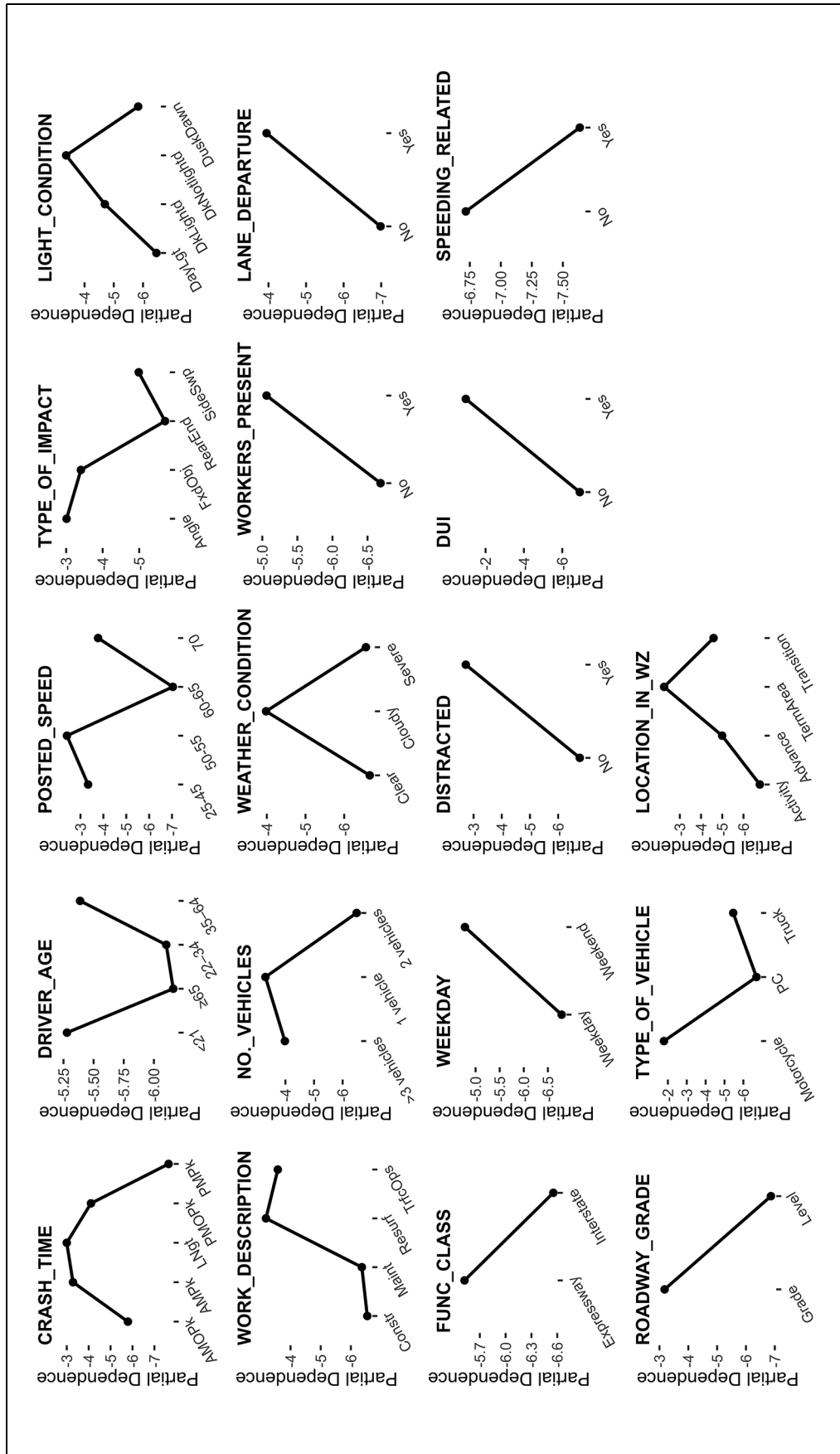


Figure 6. Partial dependency plots for predictor variable.
Note: WZ = work zone; DUI = driving under the influence.

weekdays. This trend could be attributed to weekends being characterized by less traffic congestion, therefore higher speeds than congested conditions during weekdays. In their study, Osman et al. attributed the higher severity of weekend work zone crashes to the higher likelihood of speeding among motorists with the assumption of nonoperational work zones (19). Crashes occurring in these situations are therefore likely to lead to forceful impacts resulting in severe injuries or fatalities. However, studies that analyzed non-work-zone crashes indicated that more severe crashes occurred on weekdays rather than weekends (54, 55).

Concerning the time of the crash, the partial dependency plot indicated that morning peak hours and late-night hours had higher injury risk. Severe crashes during morning peak hours could be attributed to increased traffic volumes during this period or aggressive driving by those trying to reach their destinations. Such behavior could be attributed to drivers' lack of compliance, resulting in more severe injuries in the case of a crash. Chen and Jovanis (56) and Duddu et al. (57) identified similar findings. In their studies, Chen and Jovanis (56) and Duddu et al. (57) identified that late-night and early morning crashes had a higher probability of fatal and serious injuries. Other studies have documented similar findings (58).

Environmental Factors

The findings indicated that crashes occurring in dark, unlighted conditions had the highest probability of resulting in severe injuries and could be attributed to poor visibility. Additionally, nighttime crashes may be characterized by fatigue or circadian rhythm disruptions among drivers, leading to slower reaction times, which could result in injuries when crashes occur (59). The impact could be further heightened in freeway work zones, where the combination of higher speed limits, work zone equipment, and temporary traffic patterns could lead to severe injuries. Several studies have highlighted the safety concerns associated with dark or unlighted conditions along work zones. For example, Madarshahian et al. demonstrated that crashes occurring on higher speed roads (>60 mph) during the night are 2.6 times more likely to result in serious injuries compared with those occurring in the daytime (60). Other studies have identified similar trends (21, 61–63). The findings indicate the complexity involved in driving along work zones, especially on freeways characterized by higher speed limits. From the weather condition perspective, the partial dependency plot indicated that crashes occurring in cloudy weather had a higher probability of resulting in severe injury crashes compared with those under severe weather. This counterintuitive

finding may suggest that drivers tend to be more aware of the potential hazards of severe weather and exercise more caution—a hypothesis supported by previous studies (19, 64).

Roadway Characteristics

The analysis of roadway characteristics revealed that crashes occurring on high-speed freeways had the highest probability of resulting in K + A crashes compared with those on lower-speed facilities. This finding is primarily attributed to the increased kinetic energy dissipated when a collision occurs on high-speed facilities, leading to more serious injuries and even fatalities. In addition, higher speeds along freeways lead to a reduction in the time drivers have to react, which, combined with sudden traffic pattern changes and obstacles along work zones, could lead to severe injuries. These findings are consistent with numerous studies highlighting safety concerns around high speeds, emphasizing the need for more speed management measures, especially along work zones (20, 65). The findings also indicated that work zone crashes occurring on expressways were more likely to result in fatalities and serious injuries compared with those on interstates. The findings also showed that freeway work zone-related crashes involving three or more vehicles were observed to have the highest probability of resulting in fatalities and serious injuries compared to those that involve two vehicles. The involvement of three or more vehicles is associated with increased kinetic energy and reduced maneuvering space, which would result in more severe crashes. Figure 6 indicates that crashes occurring along work zones involving traffic operations had the highest probability of resulting in serious injuries. Higher injury severity identified in traffic operations work zones could be attributed to dynamic traffic control measures, increased complexity, and changing traffic patterns.

Work Zone Characteristics

The findings indicated that crashes occurring along work zones related to resurfacing activities and traffic operations were most likely to result in fatal or incapacitating injuries compared with other types of work zone activities. Crashes occurring along construction work zones were observed to have the lowest probability of resulting in fatal or severe injuries. One explanation for this trend is the possibility that construction work zones are usually long-term and therefore more effort and safety measures may be deployed, leading to fewer severe crashes. On the contrary, traffic operations work zones are usually short-term, as they mainly involve setting up signs, lights, or other short-term countermeasures aimed at improving

traffic flow. Such short-term activities and could lead to fewer safety measures being put in place, leading to a higher risk of severe injuries.

Previous studies found similar trends, for instance, Rouphail et al. identified that the crash rate was higher at long-term construction sites, but the rate of fatal and serious injury severity was high at short-term sites (66). One possible explanation is related to speed discrepancy, that is, the difference between the normal roadway speed limit and the reduced speed limit within the work zone. More instances of speed discrepancy were found in short-term work zones, whereas long-term work zones were observed to have less speed discrepancy. Another study stated that the short-term nature of utility work zones (which includes traffic operations work zones) is always characterized with longer setting up times compared with the actual work zone. As a result, using a complex setup process might actually increase the risk to workers rather than reduce it (67).

Driver Behavior

Among the factors associated with driver behavior and characteristics, driver age emerged as the most important variable. Drivers aged between 35 and 64 showed the highest probability of incurring severe injury crashes compared with other age groups. This could be attributed to the distribution of drivers, that is, most drivers fall within this age category. Drivers aged 21 and below showed the second highest probability of incurring fatal and severe injuries. Young drivers are more prone to severe crashes owing to various factors, including inexperience, risk-taking behaviors, and sometimes lower adherence to traffic regulations. The combination of such factors, higher speeds, and temporary traffic maneuvers found along freeway work zones could contribute to the occurrence of fatal and severe injury crashes among young drivers. This finding is consistent with several other studies (17, 68). The findings also indicated that drivers aged 65 and above face the lowest risk of sustaining severe and fatal injuries in work zone crashes. This may be attributed to their greater driving experience and more cautions driving behavior compared with other age groups. A similar study found such trends, indicating that older drivers are at less risk of crashing than younger drivers (69). Conversely, research has also indicated a higher risk of severe injury crashes among older drivers compared with other age groups (70, 71).

As expected, the results indicated that the probability of a driver incurring a severe or fatal injury in the case of a freeway work zone crash was higher for drivers driving under the influence and distracted. This observation aligns with existing knowledge as alcohol- or drug-impaired drivers experience slower reaction times and

impaired judgment. Such conditions, temporary traffic control maneuvers, and sudden speed changes along freeway work zones contribute to severe crashes. Using crash data from the state of Ohio, researchers identified that work zone crashes involving an alcohol-impaired driver are twice as likely to result in serious injuries or fatalities compared with those that do not (23). In Minnesota, alcohol-impaired drivers were determined to be 146% more likely to incur fatalities or severe injuries in the case of a work zone crash (72).

Crash Characteristics

From the variable importance plot, the type of impact was observed to be the fourth most crucial variable in determining freeway work zone crash injury severity. The findings indicated that angle crashes had the highest probability of resulting in serious injuries or fatalities compared with other types of collision. The probability decreased with fixed object crashes, rear-end crashes, and sideswipe crashes. The higher injury likelihood may be attributed to the nature of angle crashes, which usually involve any of the passengers/driver being hit from the side, which could result in serious injuries or even fatalities. A study by Koilada et al. indicated similar trends, with angle collisions having the highest odds ratio among all other types of collision in both advance warning and transition areas within the work zone (73). Crashes with fixed objects at work zones could result from stationary equipment or delineator devices along work zones. Collision with such objects could lead to severe injuries, especially when combined with high speeds along freeways. Madarshahian et al. also identified that higher injury severity was associated with vehicle collisions on fixed objects along work zones (60).

Crashes involving lane departure along freeway work zones were found to have a higher probability of fatalities and severe injuries than those that did not. The influence of lane departure on injury severity may be attributed to the associated risks, including the potential for vehicles to collide with fixed objects or vehicles along work zones, and rollovers. Previous studies have identified that lane-departure-related crashes are among the deadliest collision types (74).

The presence of workers along freeway work zones was identified as influencing the occurrence of fatal and severe injury crashes compared with when workers are absent. With higher speed limits along freeways, crashes that occur in work zones with workers present tend to be more severe because of the added vulnerability of the workers. In addition to that, workers' presence could be a source of distraction among drivers. Distracted driving and high speeds along work zones could contribute to

the heightened risk of severe injuries in the event of a crash. Multiple previous studies support these conclusions (16, 75).

Conclusions and Recommendations

Despite the safety concerns over work zones along our roadways, work zones remain highly prevalent given the need for continuous capacity improvement, highway construction, and the aging highway infrastructure. Abrupt speed changes, difficulty in maneuvering, and changes in traffic management in high-speed freeway work zones are especially hazardous, leading to injuries when crashes occur. This study aimed to understand the contributing factors to work zone crashes that occur along freeways in Florida. Work zone data from FDOT's work zone archive were merged with crash data from Signal Four Analytics using crash time and location. This approach enhanced the accuracy and relevance of the analysis and has the potential to offer valuable insights on Florida's work zone safety performance.

Pearson's correlation coefficient and the Boruta algorithm were used to find the collinearity and importance among the predictor variables, respectively. Six different models were developed, including the K-NN, RF, SVM, XGBoost, MLR, and OLR. The area under the receiver operating characteristic and the overall accuracy were used to determine the best-performing model. The RF model demonstrated the highest overall accuracy, therefore it was chosen for further analysis. Time of crash occurrence, type of impact, driver age, and posted speed limit were observed to be the most critical features in predicting crash injury severity along Florida freeways. Partial dependency plots were used to interpret the Random Forest model for the fatal and severe injury crashes. Findings indicated that crashes occurring along freeway work zones during the morning peak hour and late at night had the highest probability of resulting in fatalities and serious injuries. Angle and fixed object collisions are also expected to result in fatalities and severe injuries in the case of a freeway work zone crash. The study findings showed that drivers aged between 35 and 64 were at the highest risk of incurring serious injuries, followed by drivers younger than 21 years. Additionally, crashes at work zones with posted limits between 50 and 55 mph were at the highest risk of serious injuries, followed by those that occur along segments with 70 mph speed limits.

The findings highlighted the need for lighting along Florida's freeway work zones to improve visibility and enhance compliance during dark conditions. In addition, the need for speed regulation was emphasized along Florida's freeway work zones to reduce the extent of injuries when crashes occur. Instead of sudden speed

limit changes, gradual changes could be used to prepare drivers when approaching work zones on freeways. The findings also suggest that transportation agencies and contractors should ensure proper traffic management practices, especially when workers are present. In addition, transportation agencies could develop educational programs that inform or remind drivers of the need to drive carefully, especially along work zones. Enhanced vehicle technology could also eliminate distractions and prevent driving under the influence. The findings from this study could be utilized by other state agencies to reduce work zone crash severity and improve overall freeway work zone safety.

Study Limitations and Future Studies

This study analyzed 1,258 of 14,957 crashes that occurred between 2018 and 2022 along freeway work zones in Florida. This reduced sample size was a result of missing work zone data, which were unavailable for more than 13,000 freeway crashes, therefore limiting the dataset. This limitation might have had an impact on the robustness and generalizability of the findings. Future studies would benefit from more comprehensive work zone data collection to allow for comprehensive analyses of freeway work zone crash data. Additionally, studies would benefit from analyzing additional work zone crash features such as the number of lanes closed during the crash occurrence, the traffic control devices present along the work zone, the presence of enforcement, the traffic volumes at times of crash occurrence, the specific object hit along the work zone. Including the range of work zone-specific variables would significantly improve the ability to understand contributing patterns and develop targeted countermeasures, reducing crash rates, injury severity, and improving overall work zone safety.

Acknowledgment

The authors thank the Florida Department of Transportation for providing access to the work zone data and crash data.

Author Contributions

The authors confirm contribution to the paper as follows: study conception and design: H. Shita, P. Alluri; data collection: H. Shita, P. Alluri; analysis and interpretation of results: H. Shita, H. M. Nayem, P. Alluri; draft manuscript preparation: H. Shita, P. Alluri, H. M. Nayem. All authors reviewed the results and approved the final version of the manuscript.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The authors disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This work was partially supported by the Florida Department of Transportation who provided access to the Florida state work zone data and crash data.

ORCID iDs

Hellen Shita  <https://orcid.org/0000-0003-2697-4199>

HM Nayem  <https://orcid.org/0000-0002-6431-0636>

Priyanka Alluri  <https://orcid.org/0009-0003-1998-8661>

Data Accessibility Statement

The data that support the findings of this study are available from the corresponding author on reasonable request with approval from the Florida Department of Transportation.

References

1. FHWA. *Manual on Uniform Traffic Control Devices*. Transportation Research Board, Washington, D.C., 2009.
2. Zhao, L., S. I. Chien, and B. Du. Optimizing Work-Zone Schedule with Floating Car Data Considering Traffic Diversion and Managed Lanes. *Journal of Transportation Engineering, Part A: Systems*, Vol. 145, No. 1, 2019, pp. 1–10. <https://doi.org/10.1061/jtepbs.0000202>.
3. Pesti, G., P. Wiles, R. Long, K. Cheu, P. Songchitruksa, J. Shelton, and S. Cooner. Traffic Control Strategies for Congested Freeways and Work Zones. No. FHWA/TX-08/0-5326-2. Texas Transportation Institute, 2008.
4. Dias, I. Work Zone Crash Analysis and Modeling to Identify the Factors Affecting Crash Severity and Frequency. Kansas State University, 2015.
5. Petzoldt, T., P. Rossner, C. Mair, A. C. Bullinge, and J. F. Krems. How Does Left Lane Width in Motorway Work Zones Affect Driver Behaviour and Perception? *Advances in Transportation Studies*, Vol. 1, No. 1, 2016, pp. 51–58.
6. Mohammed, H. J., Y. I. Chang, and S. D. Schrock. Factors Associated with Work Zone Crashes. *Transportation Research Record: Journal of the Transportation Research Board*, 2023. 2677: 224–235.
7. NHTSA. National Highway Traffic Safety Administration Fatality Analysis Reporting System (FARS). <https://www.nhtsa.gov/research-data/fatality-analysis-reporting-system-fars>.
8. Yang, H., K. Ozbay, O. Ozturk, and K. Xie. Work Zone Safety Analysis and Modeling: A State-of-the-Art Review. *Traffic Injury Prevention*, Vol. 16, No. 4, 2015, pp. 387–396. <https://doi.org/10.1080/15389588.2014.948615>.
9. Sweet, M. Does Traffic Congestion Slow the Economy? *Journal of Planning Literature*, Vol. 26, No. 4, 2011, pp. 391–404. <https://doi.org/10.1177/0885412211409754>.
10. American Road & Transportation Builders Association (ARTBA). *Work Zone Safety Information Clearinghouse*. <https://workzonesafety.org/work-zone-data/work-zone-traffic-crash-trends-and-statistics/>. Accessed January 31, 2023.
11. Yang, H., K. Ozbay, O. Ozturk, and M. Yildirimoglu. Modeling Work Zone Crash Frequency by Quantifying Measurement Errors in Work Zone Length. *Accident Analysis & Prevention*, Vol. 55, 2013, pp. 192–201. <https://doi.org/10.1016/j.aap.2013.02.031>.
12. Choudhary, P., M. Imprialou, N. R. Velaga, and A. Choudhary. Impacts of Speed Variations on Freeway Crashes by Severity and Vehicle Type. *Accident Analysis & Prevention*, Vol. 121, 2018, pp. 213–222. <https://doi.org/10.1016/j.aap.2018.09.015>.
13. Norman, O. Braking Distance of Vehicles from High Speed. *Public Roads*, Vol. 27, 1952, pp. 159–169.
14. La Torre, F., L. Domenichini, and A. Nocentini. Effects of Stationary Work Zones on Motorway Crashes. *Safety Science*, Vol. 92, 2017, pp. 148–159. <https://doi.org/10.1016/j.ssci.2016.10.008>.
15. Xie, Y., K. Zhao, and N. Huynh. Analysis of Driver Injury Severity in Rural Single-Vehicle Crashes. *Accident Analysis & Prevention*, Vol. 47, 2012, pp. 36–44. <https://doi.org/10.1016/j.aap.2011.12.012>.
16. Sze, N. N., and Z. Song. Factors Contributing to Injury Severity in Work Zone Related Crashes in New Zealand. *International Journal of Sustainable Transportation*, Vol. 13, No. 2, 2019, pp. 148–154. <https://doi.org/10.1080/15568318.2018.1452083>.
17. Zhang, K., and M. Hassan. Identifying the Factors Contributing to Injury Severity in Work Zone Rear-End Crashes. *Journal of Advanced Transportation*, Vol. 2019, 2019, p. 4126102. <https://doi.org/10.1155/2019/4126102>.
18. Ghasemzadeh, A., and M. M. Ahmed. Exploring Factors Contributing to Injury Severity at Work Zones Considering Adverse Weather Conditions. *IATSS Research*, Vol. 43, No. 3, 2019, pp. 131–138. <https://doi.org/10.1016/j.iatssr.2018.11.002>.
19. Osman, M., R. Paleti, and S. Mishra. Analysis of Passenger-Car Crash Injury Severity in Different Work Zone Configurations. *Accident Analysis & Prevention*, Vol. 111, 2018, pp. 161–172. <https://doi.org/10.1016/j.aap.2017.11.026>.
20. Khorashadi, A., D. Niemeier, V. Shankar, and F. Mannering. Differences in Rural and Urban Driver-Injury Severities in Accidents Involving Large-Trucks: An Exploratory Analysis. *Accident Analysis & Prevention*, Vol. 37, No. 5, 2005, pp. 910–921. <https://doi.org/10.1016/j.aap.2005.04.009>.
21. Osman, M., R. Paleti, S. Mishra, and M. M. Golias. Analysis of Injury Severity of Large Truck Crashes in Work Zones. *Accident Analysis & Prevention*, Vol. 97, 2016, pp. 261–273. <https://doi.org/10.1016/j.aap.2016.10.020>.
22. Garber, N. J., and M. Zhao. Crash Characteristics at Work Zones. No. VTRC 02-R12. Virginia Transportation Research Council, 2002.
23. Katta, V. *Development of Crash Severity Model for Predicting Risk Factors in Work Zones for Ohio*. The University of Toledo, OH, 2013.
24. Brinkman, J., and J. Lin. Freeway Revolts! The Quality of Life Effects of Highways. *Review of Economics and Statistics*, Vol. 106, No. 5, 2024, pp. 1268–1284. https://doi.org/10.1162/rest_a_01244.
25. Kitali, A. E., P. Alluri, T. Sando, H. Haule, E. Kidando, and R. Lentz. Likelihood Estimation of Secondary Crashes

- Using Bayesian Complementary Log-Log Model. *Accident Analysis & Prevention*, Vol. 119, 2018, pp. 58–67. <https://doi.org/10.1016/j.aap.2018.07.003>.
26. Kobelo, D., V. Patrangenaru, and R. Mussa. Safety Analysis of Florida Urban Limited Access Highways with Special Focus on the Influence of Truck Lane Restriction Policy. *Journal of Transportation Engineering*, Vol. 134, No. 7, 2008, pp. 297–306. [https://doi.org/10.1061/\(ASCE\)0733-947X\(2008\)134:7\(297\)](https://doi.org/10.1061/(ASCE)0733-947X(2008)134:7(297)).
 27. Dissanayake, S., and U. Roy. Crash Severity Analysis of Single Vehicle Run-off-Road Crashes. *Journal of Transportation Technologies*, Vol. 4, No. 1, 2014, pp. 1–10. <https://doi.org/10.4236/jtts.2014.41001>.
 28. Kursa, M., and W. Rudnicki. Feature Selection with the Boruta Package. *Journal of Statistical Software*, Vol. 36, No. 11, 2010, pp. 1–13.
 29. Alnami, H. M., I. Mahgoub, and H. Al-Najada. Highway Accident Severity Prediction for Optimal Resource Allocation of Emergency Vehicles and Personnel. *Proc., 2021 IEEE 11th Annual Computing and Communication Workshop and Conference, CCWC 2021*, IEEE, New York, 2021, pp. 1231–1238. <https://doi.org/10.1109/CCWC51732.2021.9376155>.
 30. Anand, N., R. Sehgal, S. Anand, and A. Kaushik. Feature Selection on Educational Data Using Boruta Algorithm. *International Journal of Computational Intelligence Studies*, Vol. 10, No. 1, 2021, p. 27. <https://doi.org/10.1504/ijcistudies.2021.113826>.
 31. Leong, L. K., and A. A. Abdullah. Prediction of Alzheimer's Disease (AD) Using Machine Learning Techniques with Boruta Algorithm as Feature Selection Method. *Journal of Physics: Conference Series*, Vol. 1372, No. 1, 2019, p. 012065. <https://doi.org/10.1088/1742-6596/1372/1/012065>.
 32. Joyce, B., S. Salaja, and E. Rajsingh. *Classification of Road Accidents Using SVM and KNN*. International Conference on Artificial Intelligence and Data Engineering. Springer Nature Singapore, Singapore, 2019, pp. 27–41.
 33. Zhang, J., Z. Li, Z. Pu, and C. Xu. Comparing Prediction Performance for Crash Injury Severity Among Various Machine Learning and Statistical Methods. *IEEE Access*, Vol. 6, 2018, pp. 60079–60087. <https://doi.org/10.1109/ACCESS.2018.2874979>.
 34. Breiman, L. Random Forests. *Machine Learning*, Vol. 45, No. 1, 2001, pp. 5–32. <https://doi.org/10.1023/A:1010933404324>.
 35. Wu, J., H. Xu, Y. Zheng, Y. Zhang, B. Lv, and Z. Tian. Automatic Vehicle Classification Using Roadside LiDAR Data. *Transportation Research Record: Journal of the Transportation Research Board*, 2019. 2673: 153–164.
 36. Yu, B., Y. Chen, and S. Bao. Quantifying Visual Road Environment to Establish a Speeding Prediction Model: An Examination Using Naturalistic Driving Data. *Accident Analysis & Prevention*, Vol. 129, 2019, pp. 289–298. <https://doi.org/10.1016/j.aap.2019.05.011>.
 37. Li, J., J. Guo, J. S. Wijnands, R. Yu, C. Xu, and M. Stevenson. Assessing Injury Severity of Secondary Incidents Using Support Vector Machines. *Journal of Transportation Safety and Security*, Vol. 14, No. 2, 2022, pp. 197–216. <https://doi.org/10.1080/19439962.2020.1754983>.
 38. Belgiu, M., and L. Drăgu. Random Forest in Remote Sensing: A Review of Applications and Future Directions. *ISPRS Journal of Photogrammetry and Remote Sensing*, Vol. 114, 2016, pp. 24–31. <https://doi.org/10.1016/j.isprsjprs.2016.01.011>.
 39. Li, Z., P. Liu, W. Wang, and C. Xu. Using Support Vector Machine Models for Crash Injury Severity Analysis. *Accident Analysis & Prevention*, Vol. 45, 2012, pp. 478–486. <https://doi.org/10.1016/j.aap.2011.08.016>.
 40. Delen, D., L. Tomak, K. Topuz, and E. Eryarsoy. Investigating Injury Severity Risk Factors in Automobile Crashes with Predictive Analytics and Sensitivity Analysis Methods. *Journal of Transport and Health*, Vol. 4, 2017, pp. 118–131. <https://doi.org/10.1016/j.jth.2017.01.009>.
 41. Chen, T., and C. Guestrin. XGBoost: A Scalable Tree Boosting System. *Proc., ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, San Francisco, CA, Association for Computing Machinery, New York, NY, August 13–17, 2016, pp. 785–794. <https://doi.org/10.1145/2939672.2939785>.
 42. Chen, H., H. Chen, Z. Liu, X. Sun, and R. Zhou. Analysis of Factors Affecting the Severity of Automated Vehicle Crashes Using XGBoost Model Combining POI Data. *Journal of Advanced Transportation*, Vol. 2020, 2020, p. 8881545. <https://doi.org/10.1155/2020/8881545>.
 43. Zhang, S., A. Khattak, C. M. Matara, A. Hussain, and A. Farooq. Hybrid Feature Selection-Based Machine Learning Classification System for the Prediction of Injury Severity in Single and Multiple-Vehicle Accidents. *PLoS One*, Vol. 17, 2022, pp. 1–19. <https://doi.org/10.1371/journal.pone.0262941>.
 44. Eboli, L., C. Forciniti, and G. Mazzulla. Factors Influencing Accident Severity: An Analysis by Road Accident Type. *Transportation Research Procedia*, Vol. 47, 2020, pp. 449–456. <https://doi.org/10.1016/j.trpro.2020.03.120>.
 45. Shiran, G., R. Imaninasab, and R. Khayamim. Crash Severity Analysis of Highways Based on Multinomial Logistic Regression Model, Decision Tree Techniques and Artificial Neural Network: A Modeling Comparison. *Sustainability (Switzerland)*, Vol. 13, No. 10, 2021, p. 5670. <https://doi.org/10.3390/su13105670>.
 46. Kiner, A., K. Bukuru, E. E. Mwambeleko, T. Sando, and P. Alluri. Modeling Injury Severity of Crashes Involving Golf Carts: A Case Study of The Villages, Florida. *Traffic Injury Prevention*, Vol. 25, No. 2, 2024, pp. 165–172. <https://doi.org/10.1080/15389588.2023.2291332>.
 47. Pepe, M. S. Receiver Operating Characteristic Methodology. *Journal of the American Statistical Association*, Vol. 95, No. 449, 2000, pp. 308–311. <https://doi.org/10.1080/01621459.2000.10473930>.
 48. Das, S., X. Sun, F. Wang, and C. Leboeuf. Estimating Likelihood of Future Crashes for Crash-Prone Drivers. *Journal of Traffic and Transportation Engineering (English Edition)*, Vol. 2, No. 3, 2015, pp. 145–157. <https://doi.org/10.1016/j.jtte.2015.03.003>.
 49. Novat, N., S. Mwende, H. Shita, and G. Mbiaji. Predicting Drivers' Injury Occurrence in Vehicle Crashes Occurring at Horizontal Curve and Grade Combination Segments. *ITE Journal (Institute of Transportation Engineers)*, Vol. 92, No. 8, 2022, pp. 41–50.

50. Friedman, J. H. Greedy Function Approximation: A Gradient Boosting Machine. *The Annals of Statistics*, Vol. 29, No. 5, pp. 1189–1232.
51. Li, J., J. Liu, P. Liu, and Y. Qi. Analysis of Factors Contributing to the Severity of Large Truck Crashes. *Entropy*, Vol. 22, No. 11, 2020, pp. 1–21. <https://doi.org/10.3390/e22111191>.
52. Danesh, T., R. Ouaret, and P. Floquet. Interpretability in Machine Learning Predictions: Case of Random Forest Regression Using Partial Dependence Plots. In 18ème congrès de la Société Française de Génie des Procédés, Toulouse, France, 2022.
53. Kidando, E., A. E. Kitali, B. Kutela, M. Ghorbanzadeh, A. Karaer, M. Koloushani, R. Moses, E. E. Ozguven, and T. Sando. Prediction of Vehicle Occupants Injury at Signalized Intersections Using Real-Time Traffic and Signal Data. *Accident Analysis & Prevention*, Vol. 149, 2021, p. 105869. <https://doi.org/10.1016/j.aap.2020.105869>.
54. Christoforou, Z., S. Cohen, and M. G. Karlaftis. Vehicle Occupant Injury Severity on Highways: An Empirical Investigation. *Accident Analysis & Prevention*, Vol. 42, No. 6, 2010, pp. 1606–1620. <https://doi.org/10.1016/j.aap.2010.03.019>.
55. Quddus, M. A., C. Wang, and S. G. Ison. Road Traffic Congestion and Crash Severity: Econometric Analysis Using Ordered Response Models. *Journal of Transportation Engineering*, Vol. 136, No. 5, 2010, pp. 424–435. [https://doi.org/10.1061/\(ASCE\)TE.1943-5436.0000044](https://doi.org/10.1061/(ASCE)TE.1943-5436.0000044).
56. Chen, W. H., and P. P. Jovanis. Method for Identifying Factors Contributing to Driver-Injury Severity in Traffic Crashes. *Transportation Research Record: Journal of the Transportation Research Board*, 2000. 1717: 1–9.
57. Duddu, V. R., V. M. Kukkapalli, and S. S. Pulugurtha. Crash Risk Factors Associated with Injury Severity of Teen Drivers. *IATSS Research*, Vol. 43, No. 1, 2019, pp. 37–43. <https://doi.org/10.1016/j.iatssr.2018.08.003>.
58. Hao, W., C. Kamga, and D. Wan. The Effect of Time of Day on Driver's Injury Severity at Highway-Rail Grade Crossings in the United States. *Journal of Traffic and Transportation Engineering (English Edition)*, Vol. 3, No. 1, 2016, pp. 37–50. <https://doi.org/10.1016/j.jtte.2015.10.006>.
59. Chipman, M., and Y. L. Jin. Drowsy Drivers: The Effect of Light and Circadian Rhythm on Crash Occurrence. *Safety Science*, Vol. 47, No. 10, 2009, pp. 1364–1370. <https://doi.org/10.1016/j.ssci.2009.03.005>.
60. Madarshahian, M., A. Balaram, F. Ahmed, N. Huynh, C. K. A. Siddiqui, and M. Ferguson. Analysis of Injury Severity of Work Zone Truck-Involved Crashes in South Carolina for Interstates and Non-Interstates. *Sustainability (Switzerland)*, Vol. 15, No. 9, 2023, pp. 1–18. <https://doi.org/10.3390/su15091888>.
61. Wang, K., N. Shirani-bidabadi, M. Razaur Rahman Shaon, S. Zhao, and E. Jackson. Correlated Mixed Logit Modeling with Heterogeneity in Means for Crash Severity and Surrogate Measure with Temporal Instability. *Accident Analysis & Prevention*, Vol. 160, 2021, p. 106332. <https://doi.org/10.1016/j.aap.2021.106332>.
62. Islam, M. An Analysis of Motorcyclists' Injury Severities in Work-Zone Crashes with Unobserved Heterogeneity. *IATSS Research*, Vol. 46, No. 2, 2022, pp. 281–289. <https://doi.org/10.1016/j.iatssr.2022.01.003>.
63. Uddin, M., and N. Huynh. Injury Severity Analysis of Truck-Involved Crashes Under Different Weather Conditions. *Accident Analysis & Prevention*, Vol. 141, 2020, p. 105529. <https://doi.org/10.1016/j.aap.2020.105529>.
64. Khattak, A. J., and F. Targa. Injury Severity and Total Harm in Truck-Involved Work Zone Crashes. *Transportation Research Record: Journal of the Transportation Research Board*, 2004. 1877: 106–116.
65. Liu, P., and W. Fan. Analyzing Injury Severity of Rear-End Crashes Involving Large Trucks Using a Mixed Logit Model: A Case Study in North Carolina. *Journal of Transportation Safety and Security*, Vol. 14, No. 5, 2022, pp. 723–736. <https://doi.org/10.1080/19439962.2020.1812784>.
66. Roupail, N. M., Z. S. Yang, and J. Fazio. Comparative Study of Short- and Long-Term Urban Freeway Work Zones. *Transportation Research Record: Journal of the Transportation Research Board*, 1988. 1163: 4–14.
67. Datta, T. K., P. Savolainen, L. Grillo, and K. Schattler. Utility Work Zone Traffic Control. No. DTFH61-06-G-00006. Federal Highway Administration, Washington, D.C., September 2008.
68. Rami, H., R. Essam, Y. Xuedong, M. Abdel-Aty, and P. Anurag. Environmental, Driver, and Vehicle Risk Analysis for Freeway Work Zone Crashes. Institute of Transportation Engineers. ITE Journal, Vol. 78, No. 1, 2008, p. 26.
69. Weng, J., and Q. Meng. Effects of Environment, Vehicle and Driver Characteristics on Risky Driving Behavior at Work Zones. *Safety Science*, Vol. 50, No. 4, 2012, pp. 1034–1042. <https://doi.org/10.1016/j.ssci.2011.12.005>.
70. Abdel-Aty, M. Analysis of Driver Injury Severity Levels at Multiple Locations Using Ordered Probit Models. *Journal of Safety Research*, Vol. 34, No. 5, 2003, pp. 597–603. <https://doi.org/10.1016/j.jsr.2003.05.009>.
71. Adebisi, A., J. Ma, J. Masaki, and J. Sobanjo. Age-Related Differences in Motor-Vehicle Crash Severity in California. *Safety*, Vol. 5, No. 3, 2019, p. 48.
72. Osman, M., S. Mishra, R. Paleti, and M. Golias. Impacts of Work Zone Component Areas on Driver Injury Severity. *Journal of Transportation Engineering, Part A: Systems*, Vol. 145, No. 8, 2019, pp. 1–13. <https://doi.org/10.1061/jtepbs.0000253>.
73. Koilada, K., A. S. Mane, and S. S. Pulugurtha. Odds of Work Zone Crash Occurrence and Getting Involved in Advance Warning, Transition, and Activity Areas by Injury Severity. *IATSS Research*, Vol. 44, No. 1, 2020, pp. 75–83. <https://doi.org/10.1016/j.iatssr.2019.07.003>.
74. Penmetsa, P., M. Hudnall, and S. Nambisan. Potential Safety Benefits of Lane Departure Prevention Technology. *IATSS Research*, Vol. 43, No. 1, 2019, pp. 21–26. <https://doi.org/10.1016/j.iatssr.2018.08.002>.
75. McAvoy, D. S., S. Duffy, and H. S. Whiting. Simulator Study of Primary and Precipitating Factors in Work Zone Crashes. *Transportation Research Record: Journal of the Transportation Research Board*, 2011. 2258: 32–39.

The opinions, findings, and conclusions expressed in this publication are those of the authors and do not necessarily reflect the views of Florida Department of Transportation or Florida International University.